Precision Agriculture and Digital Farming

Agriculture RAG Knowledge Base • Educational reference document

Precision agriculture uses field-level information and digital tools to make agricultural decisions more targeted. Technologies may include GPS, satellite imagery, drones, sensors, weather data, geographic information systems, and machine-learning models.

A precision-agriculture workflow commonly includes data collection, cleaning, analysis, decision-making, field action, and evaluation. For example, soil-test data can be mapped to identify zones with different nutrient characteristics. Irrigation decisions can incorporate soil moisture and weather information.

Remote sensing can provide spatial information about vegetation condition, while ground measurements can help validate those observations. Sensors can monitor variables such as soil moisture, temperature, electrical conductivity, or other field conditions depending on the application.

Machine learning can support prediction and classification, but model quality depends on representative data, correct labels, suitable validation, and monitoring after deployment. A model should not be treated as automatically correct simply because it produces a numerical prediction.

The value of precision agriculture comes from connecting data to an actionable decision. A dashboard that displays maps without a useful management workflow may have limited practical value. Good systems therefore connect measurements, uncertainty, recommendations, and farmer decisions.

Reference basis

• FAO, digital agriculture and innovation resources

• General agronomic principles from FAO soil and water resources

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.